

The population and empirical Gram matrices of the product design

wafc-draft, derivation 03 (E1.4)

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Setting

The model and the notation are those of `docs/notacao.md`. The response is $Y = \sum_{\ell=1}^p \beta_{\ell}(\mathbf{U}) X_{\ell} + \varepsilon$ with $\beta_{\ell}(u) = c_{\ell} + \sum_{m=1}^q g_{\ell m}(u_m)$, $\mathbf{X} = (X_1, \dots, X_p)^{\top}$ and $\mathbf{U} = (U_1, \dots, U_q)^{\top} \in [0, 1]^q$. Each $g_{\ell m}$ is expanded in the periodized orthonormal wavelets $\{\psi_{jk} : j = 0, \dots, J-1; k = 0, \dots, 2^j-1\}$ of a compactly supported family (D2: $j_0 = 0$, the constant scaling function discarded), so that each block (ℓ, m) carries $N_J = 2^J - 1$ wavelets and the penalized dimension is $d = pqN_J$.

Write $\Psi(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, for the vector that collects the constant and the wavelets of the q modulating covariates,

$$\Psi(u) = (1, \psi_{jk}(u_1)_{jk}, \dots, \psi_{jk}(u_q)_{jk})^{\top},$$

with the wavelets of each u_m ordered by j increasing and, within a level, by k increasing. The row of the design matrix is, in the column order frozen in D12 (the p unpenalized columns X_{ℓ} first, then the blocks (ℓ, m) in lexicographic order),

$$\mathbf{Z} = \Pi (\mathbf{X} \otimes \Psi(\mathbf{U})) \in \mathbb{R}^{p+d}, \quad p + d = pK,$$

where $\mathbf{X} \otimes \Psi(\mathbf{U})$ is the Kronecker product, whose entries are $X_{\ell} \Psi_a(\mathbf{U})$ in ℓ -major order, and Π is the fixed permutation matrix that moves the p entries $X_{\ell} \cdot 1$ to the front. The population and empirical Gram matrices are

$$\Sigma = \mathbb{E}[\mathbf{Z}\mathbf{Z}^{\top}], \quad \hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i \mathbf{Z}_i^{\top},$$

and, for the $\mathbf{U}$ part alone, $\Sigma_{\Psi} = \mathbb{E}[\Psi(\mathbf{U})\Psi(\mathbf{U})^{\top}] \in \mathbb{R}^{K \times K}$. Since Π is a permutation, Σ and $\Pi^{\top} \Sigma \Pi$ have the same spectrum, and every statement below about eigenvalues, restricted eigenvalues or compatibility constants is invariant to the column order; the order matters only for the bookkeeping of E2.1.

For a support $S \subseteq \{1, \dots, p+d\}$ with $|S| = s_0$ and a vector $\delta \in \mathbb{R}^{p+d}$, the compatibility constant of a symmetric matrix A on S is, as in Bühlmann and van de Geer (2011, Section 6.2),

$$\phi_0^2(S; A) = \min \left\{ s_0 \delta^{\top} A \delta : \|\delta_S\|_1 = 1, \|\delta_{S^c}\|_1 \leq 3 \right\},$$

and the trivial bound $\delta^{\top} A \delta \geq \lambda_{\min}(A) \|\delta\|_2^2 \geq \lambda_{\min}(A) \|\delta_S\|_1^2 / s_0$ gives

$$\phi_0^2(S; A) \geq \lambda_{\min}(A) \quad \text{for every } S. \quad (1)$$

Hence a lower bound on $\lambda_{\min}$ is a compatibility (and restricted eigenvalue) bound on every support, with the same constant and no cone restriction, exactly as in the WALL (Lemma S.3 there). This is the form in which the condition is consumed by E1.5.

Assumptions

Assumption 1 (Modulating covariates). $\mathbf{U}$ takes values in $[0, 1]^q$ and has a joint Lebesgue density $p_{\mathbf{U}}$ with

$$0 < c_U \leq p_{\mathbf{U}}(u) \leq C_U < \infty \quad \text{for all } u \in [0, 1]^q.$$

Assumption 2 (Linear covariates). $\|\mathbf{X}\|_{\infty} \leq B_X$ almost surely, and there are constants $0 < \kappa_1 \leq \kappa_2 < \infty$ such that

$$\kappa_1 \leq \lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^{\top} \mid \mathbf{U}]) \leq \lambda_{\max}(\mathbb{E}[\mathbf{X}\mathbf{X}^{\top} \mid \mathbf{U}]) \leq \kappa_2 \quad \text{almost surely.}$$

Assumption 3 (Wavelet basis). The wavelets ψ_{jk} , $j \geq 0$, are the periodization to $[0, 1]$ of an orthonormal, compactly supported family, so that $\{1\} \cup \{\psi_{jk}\}$ is orthonormal in $L_2[0, 1]$ under the Lebesgue measure and $\int_0^1 \psi_{jk} = 0$. There is a constant C_{ψ} , depending only on the family, such that

$$\sup_{u \in [0, 1]} \sum_{j=0}^{J-1} \sum_{k=0}^{2^j-1} \psi_{jk}(u)^2 \leq C_{\psi} 2^J \quad \text{for every } J \geq 1. \quad (2)$$

Assumption 1 is D11 read for the joint law; see Remark 2. In Assumption 2, the constant κ_1 is the quantity that the identifiability step (E1.2) already needs; when $\mathbf{X}$ and $\mathbf{U}$ are independent it reduces to $\lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^{\top}]) \geq \kappa_1$. The bound (2) holds for every compactly supported family: at each level j , at most L periodized wavelets are nonzero at a given u , where L is the support length of the mother wavelet, and $\|\psi_{jk}\|_{\infty} = 2^{j/2} \|\psi\|_{\infty}$, so the left side is at most $L \|\psi\|_{\infty}^2 \sum_{j < J} 2^j \leq L \|\psi\|_{\infty}^2 2^J$. The numerical check finds $C_{\psi} \approx 1.35$ for Daubelets with filter size 8.

Statement

Proposition 1 (Gram matrix of the product design). *Under Assumptions 1 to 3, with $D = p + d = p(1 + qN_J)$:*

- (i) (The $\mathbf{U}$ part.) $c_U \leq \lambda_{\min}(\Sigma_{\Psi}) \leq \lambda_{\max}(\Sigma_{\Psi}) \leq C_U$, for every J .
- (ii) (Factorization under independence.) *If $\mathbf{X}$ and $\mathbf{U}$ are independent, then*

$$\begin{aligned} \Sigma &= \Pi (\mathbb{E}[\mathbf{X}\mathbf{X}^{\top}] \otimes \Sigma_{\Psi}) \Pi^{\top}, \\ \lambda_{\min}(\Sigma) &= \lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^{\top}]) \lambda_{\min}(\Sigma_{\Psi}) \geq \kappa_1 c_U, \\ \lambda_{\max}(\Sigma) &= \lambda_{\max}(\mathbb{E}[\mathbf{X}\mathbf{X}^{\top}]) \lambda_{\max}(\Sigma_{\Psi}) \leq \kappa_2 C_U. \end{aligned}$$

- (iii) (General case.) *Without independence,*

$$\kappa_1 \lambda_{\min}(\Sigma_{\Psi}) \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq \kappa_2 \lambda_{\max}(\Sigma_{\Psi}),$$

hence $\kappa_1 c_U \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq \kappa_2 C_U$. In particular $\phi_0^2(S; \Sigma) \geq \kappa_1 c_U$ for every support S , the unpenalized coordinates included, by (1).

- (iv) (Empirical version.) *Let $(\mathbf{X}_i, \mathbf{U}_i)$, $i = 1, \dots, n$, be i.i.d. copies of $(\mathbf{X}, \mathbf{U})$ and $R_J = p B_X^2 (1 + q C_{\psi} 2^J)$. Then $\|\mathbf{Z}_i\|_2^2 \leq R_J$ almost surely and, for every $t > 0$,*

$$\mathbb{P}(\|\hat{\Sigma} - \Sigma\|_{\text{op}} \geq t) \leq 2D \exp\left(-\frac{n t^2}{2R_J (\kappa_2 C_U + 2t/3)}\right).$$

Consequently, with $\gamma = \kappa_1 c_U$,

$$\mathbb{P}\left(\lambda_{\min}(\widehat{\Sigma}) \geq \frac{\gamma}{2}\right) \geq 1 - 2D \exp\left(-\frac{n\gamma^2}{8R_J(\kappa_2 C_U + \gamma/3)}\right),$$

which tends to 1 whenever $pq2^J \log(pq2^J)/n \rightarrow 0$, and on that event $\phi_0^2(S; \widehat{\Sigma}) \geq \gamma/2$ for every support S .

Remark 1 (Cone-restricted alternative for the empirical step). Part (iv) transfers the full smallest eigenvalue, which is what the sieve regime $2^{J_n} \asymp n^{1/(2s'+1)}$ of E1.6 makes available at no cost: there $pq2^J \log(pq2^J)/n \rightarrow 0$ as soon as pq grows slower than $n^{2s'/(2s'+1)}$ up to logarithms. If pq were larger, the compatibility constant alone can still be transferred through the entrywise deviation: by Bühlmann and van de Geer (2011, Corollary 6.8, Section 6.12), $\|\widehat{\Sigma} - \Sigma\|_\infty \leq \phi_0^2(S; \Sigma)/(32s_0)$ implies $\phi_0^2(S; \widehat{\Sigma}) \geq \phi_0^2(S; \Sigma)/2$, and a union bound over the D^2 entries, each bounded by $B_X^2 C'_\psi 2^J$ with variance at most $B_X^4 C_U C'_\psi 2^J$, gives $\|\widehat{\Sigma} - \Sigma\|_\infty = O_p(2^{J/2} \sqrt{\log D/n} + 2^J \log D/n)$; the requirement becomes $s_0^2 2^J \log(pq2^J)/n \rightarrow 0$, which involves pq only through the logarithm. This is the " $s_0 \log(pqN_J)$ " form anticipated in the plan; it is not needed for the rates of E1.6 and is recorded for the case $pq \gg n^{1/2}$.

Remark 2 (Joint versus marginal density). Part (i) is where the lower bound c_U on the density enters, and it is the joint density on $[0, 1]^q$ that is needed. Marginal bounds are not enough: if $U_2 = U_1$ almost surely, each marginal is uniform, but $\psi_{jk}(U_1) - \psi_{jk}(U_2) \equiv 0$ and Σ_Ψ is singular. The joint lower bound is thus the "no concavity among the modulating covariates" condition of `proposta-metodo.md`, §4, item 2, made quantitative: it rules out one U_m being a function of the others. The WALL imposes the lower bound on $\lambda_{\min}$ of the population Gram directly (its Assumption 5.7) and lets the density lower bound play "only a motivating role"; here the same bound is derived from the density, with the explicit constant c_U .

Remark 3 (What $\kappa_1 > 0$ excludes). Assumption 2 fails exactly when some nontrivial linear combination of $X_1, \dots, X_p$ is almost surely a function of $\mathbf{U}$ on a set of positive probability. The canonical case is $X_1 \equiv 1$ and $X_2 = h(U_1)$: then $\mathbb{E}[\mathbf{X}\mathbf{X}^\top \mid \mathbf{U}]$ has rank one, $\kappa_1 = 0$, and the population Gram degenerates as J grows, because $h(u_1)\psi_{jk}(u_1)$ is approached by the span of $\{1, \psi_{j'k'}(u_1)\}$. The numerical check exhibits this with $h(u) = u - 1/2$: $\lambda_{\min}(\Sigma)$ is 4×10^{-4} at $J = 2$ and below 10^{-4} at $J \geq 3$. Nothing in the proposition is lost by allowing dependence between $\mathbf{X}$ and $\mathbf{U}$ as long as $\kappa_1 > 0$; independence buys only the exact factorization of part (ii), not a better constant.

Proof

Part (i). For $a \in \mathbb{R}^K$ write $h_a(u) = a^\top \Psi(u) = a_0 + \sum_{m=1}^q h_{a,m}(u_m)$ with $h_{a,m} = \sum_{jk} a_{m,jk} \psi_{jk}$. Under Assumption 3 the functions $1, \psi_{jk}(u_1), \dots, \psi_{jk}(u_q)$ are orthonormal in $L_2([0, 1]^q, du)$: within one coordinate this is the orthonormality of the periodized system; across coordinates $\int \psi_{jk}(u_m) \psi_{j'k'}(u_{m'}) du = \int_0^1 \psi_{jk} \int_0^1 \psi_{j'k'} = 0$ for $m \neq m'$, and $\int \psi_{jk}(u_m) du = 0$, both because every wavelet integrates to zero. Hence $\int_{[0,1]^q} h_a(u)^2 du = \|a\|_2^2$, and by Assumption 1

$$c_U \|a\|_2^2 \leq \mathbb{E}[h_a(\mathbf{U})^2] = \int h_a(u)^2 p_{\mathbf{U}}(u) du \leq C_U \|a\|_2^2.$$

Since $a^\top \Sigma_\Psi a = \mathbb{E}[h_a(\mathbf{U})^2]$, this is (i).

Part (iii) first. Let $\delta \in \mathbb{R}^{pK}$ and reshape it as $\Delta \in \mathbb{R}^{p \times K}$, row ℓ holding the K coefficients attached to X_ℓ , so that $\delta = \text{vec}(\Delta^\top)$ and $(\mathbf{X} \otimes \Psi(\mathbf{U}))^\top \delta = \mathbf{X}^\top \Delta \Psi(\mathbf{U})$. By the tower property,

$$\delta^\top \Pi^\top \Sigma \Pi \delta = \mathbb{E}[(\mathbf{X}^\top \Delta \Psi(\mathbf{U}))^2] = \mathbb{E}[\Psi(\mathbf{U})^\top \Delta^\top \mathbb{E}[\mathbf{X}\mathbf{X}^\top | \mathbf{U}] \Delta \Psi(\mathbf{U})].$$

Assumption 2 bounds the inner quadratic form between $\kappa_1 \|\Delta \Psi(\mathbf{U})\|_2^2$ and $\kappa_2 \|\Delta \Psi(\mathbf{U})\|_2^2$ almost surely, and $\mathbb{E} \|\Delta \Psi(\mathbf{U})\|_2^2 = \sum_{\ell=1}^p \Delta_\ell \Sigma_\Psi \Delta_\ell^\top = \delta^\top (I_p \otimes \Sigma_\Psi) \delta$, whose value lies between $\lambda_{\min}(\Sigma_\Psi) \|\delta\|_2^2$ and $\lambda_{\max}(\Sigma_\Psi) \|\delta\|_2^2$. Therefore

$$\kappa_1 \lambda_{\min}(\Sigma_\Psi) \|\delta\|_2^2 \leq \delta^\top \Pi^\top \Sigma \Pi \delta \leq \kappa_2 \lambda_{\max}(\Sigma_\Psi) \|\delta\|_2^2,$$

and since Π is orthogonal this is the first display of (iii); part (i) gives the second. The compatibility statement is (1).

Part (ii). Under independence $\mathbb{E}[\mathbf{X}\mathbf{X}^\top | \mathbf{U}] = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$, and the same computation gives, for every δ , $\delta^\top \Pi^\top \Sigma \Pi \delta = \mathbb{E}[\Psi(\mathbf{U})^\top \Delta^\top \mathbb{E}[\mathbf{X}\mathbf{X}^\top] \Delta \Psi(\mathbf{U})] = \delta^\top (\mathbb{E}[\mathbf{X}\mathbf{X}^\top] \otimes \Sigma_\Psi) \delta$; two symmetric matrices with the same quadratic form are equal, which is the factorization. The eigenvalues of a Kronecker product of symmetric matrices are the products of the eigenvalues, so $\lambda_{\min}(\Sigma) = \lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^\top]) \lambda_{\min}(\Sigma_\Psi)$ and likewise for $\lambda_{\max}$. Finally $\lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^\top]) \geq \kappa_1$ because $\lambda_{\min}$ is concave and $\mathbb{E}[\mathbf{X}\mathbf{X}^\top] = \mathbb{E}[\mathbb{E}[\mathbf{X}\mathbf{X}^\top | \mathbf{U}]]$ (Jensen); the upper bound is the same argument with $\lambda_{\max}$ convex. (Independence is used only through $\mathbb{E}[\mathbf{X}\mathbf{X}^\top | \mathbf{U}] = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$; that weaker condition suffices for (ii).)

Part (iv). The row norm: $\|\mathbf{Z}_i\|_2^2 = \|\mathbf{X}_i\|_2^2 \|\Psi(\mathbf{U}_i)\|_2^2 \leq p B_X^2 (1 + q C_\psi 2^J) = R_J$ by (2) applied to each of the q coordinates. The rest transposes Step 2b of the WALL's Lemma S.3 to unweighted summands. Set $A_i = (\mathbf{Z}_i \mathbf{Z}_i^\top - \Sigma)/n$, independent, centered, symmetric, with $\hat{\Sigma} - \Sigma = \sum_i A_i$. Their range: $\|\mathbf{Z}_i \mathbf{Z}_i^\top\|_{\text{op}} = \|\mathbf{Z}_i\|_2^2 \leq R_J$ and $\|\Sigma\|_{\text{op}} \leq \mathbb{E} \|\mathbf{Z}\mathbf{Z}^\top\|_{\text{op}} \leq R_J$, so $\|A_i\|_{\text{op}} \leq 2R_J/n$. Their variance: $\mathbb{E}[(\mathbf{Z}\mathbf{Z}^\top - \Sigma)^2] = \mathbb{E}[\|\mathbf{Z}\|_2^2 \mathbf{Z}\mathbf{Z}^\top] - \Sigma^2 \preceq R_J \Sigma$, hence $\|\sum_i \mathbb{E} A_i^2\|_{\text{op}} \leq R_J \lambda_{\max}(\Sigma)/n \leq R_J \kappa_2 C_U/n$ by (iii). The matrix Bernstein inequality (Tropp, 2012, Theorem 1.4, applied to $\pm \sum_i A_i$) gives

$$\mathbb{P}(\|\hat{\Sigma} - \Sigma\|_{\text{op}} \geq t) \leq 2D \exp\left(-\frac{t^2/2}{R_J \kappa_2 C_U/n + (2R_J/n)(t/3)}\right) = 2D \exp\left(-\frac{n t^2}{2R_J(\kappa_2 C_U + 2t/3)}\right).$$

Take $t = \gamma/2$. On the complement event Weyl's inequality gives $\lambda_{\min}(\hat{\Sigma}) \geq \lambda_{\min}(\Sigma) - \gamma/2 \geq \gamma/2$, using (iii). The exponent is $-cn/R_J$ with $c = \gamma^2/\{8(\kappa_2 C_U + \gamma/3)\}$ fixed, and $R_J \asymp pq2^J$, $\log(2D) \asymp \log(pq2^J)$; the probability bound tends to zero exactly when $n/(pq2^J) - \log(pq2^J) \rightarrow \infty$, which is implied by $pq2^J \log(pq2^J)/n \rightarrow 0$ (the weaker $n/(pq2^J) \rightarrow \infty$ alone does not control the prefactor $2D$, as the WALL notes). The compatibility statement on the event is again (1). $\square$

What this does not cover

- **The interval boundary.** With `boundary = "interval"` (Cohen, Daubechies and Vial, 1993) the basis is still orthonormal on $[0, 1]$, so part (i) and the proof go through with N_J and C_ψ adjusted, *provided the constant is represented once*. The CDV scaling functions at level j_0 reproduce constants, so keeping them together with the unpenalized X_ℓ makes Σ singular; the design must drop one scaling function per block, or impose $\int g_{\ell m} = 0$ explicitly. This is E1.2's question, not E1.4's; the proposition assumes the periodic design of D2. Numerically, the installed `WaveBased` requires $j_0 \geq 6$ for the interval boundary with filter size 20, so the option is out of reach at $J = 3$.

- **Unbounded $\mathbf{X}$.** Part (iv) uses $\|\mathbf{X}\|_\infty \leq B_X$ through the range R_J . Sub-Gaussian $\mathbf{X}$ would need a truncation step (or the Bernstein inequality for sub-exponential matrices); parts (i) to (iii) need only second moments.
- **The empirical rescaling of $\mathbf{U}$.** D11 puts the monotone transformation to $[\epsilon, 1 - \epsilon]$ outside the theory. If the practical design uses $\hat{F}_m(U_m)$ instead of U_m , the Gram changes by a random reparametrization and the concentration step does not apply as stated.
- **Growing p .** The constants κ_1, κ_2, B_X are assumed uniform in n ; if $p = p_n$, Assumption 2 must hold uniformly and R_J carries the factor p . Nothing here treats $p \gg n$.
- **Sharpness at small n .** The bound in (iv) is not informative at $n = 200$, $D = 30$ ($R_J \log D/n \approx 1.6$): the check finds $\lambda_{\min}(\hat{\Sigma})$ at about 0.4 to 0.5 of $\lambda_{\min}(\Sigma)$, the Marchenko–Pastur order $(1 - \sqrt{D/n})^2 \approx 0.38$, while the compatibility constant on a support of size 7 is at about 0.65 to 0.75 of its population value. The restricted quantity is what E1.5 uses, and it is healthier than the full eigenvalue at the sample sizes of E2.

Numerical check

derivations/check/03-desenho-produtos.R (Daubelets, filter size 8; $n = 200$; $J = 3$; $p = 2$ with $X_1 \equiv 1$; $q = 2$; $\mathbf{U}$ uniform on $[0, 1]^2$, so $c_U = C_U = 1$ and $\Sigma_{\Psi} = I_{15}$; $D = 30$). The population Gram is computed by midpoint quadrature on a 400×400 grid with $\mathbb{E}[\mathbf{X}\mathbf{X}^\top | \mathbf{U} = u]$ in closed form. Scenario A has $X_2 = 1/2 + \eta$, $\eta \sim \text{Unif}[-1, 1]$ independent of $\mathbf{U}$; scenario B has $X_2 = (U_1 - 1/2) + \eta$, so $\kappa_1 = 0.2503$ and $\kappa_2 = 1.3318$ (extremes of the pointwise eigenvalues over $u_1 \in [0, 1]$); scenario C has $X_2 = U_1 - 1/2$, so $\kappa_1 = 0$. What it printed on 2026-09-18:

- Orthonormality of $\{1, \psi_{jk}(u_m)\}$ on the grid to 3×10^{-5} (quadrature error); $C_\psi(J)$ equal to 1.105, 1.238, 1.300, 1.331 and 1.347 for $J = 2, \dots, 6$.
- (ii) Scenario A: Σ equals the permuted Kronecker product to 7.6×10^{-13} ; $\lambda_{\min}(\Sigma) = 0.2500 = 0.2500 \times 1$, $\lambda_{\max}(\Sigma) = 1.3333 = 1.3333 \times 1$.
- (iii) Scenario B: $\lambda_{\min}(\Sigma) = 0.2747 \geq \kappa_1 = 0.2503$; $\lambda_{\max}(\Sigma) = 1.2271 \leq \kappa_2 = 1.3318$; $\lambda_{\min}(\mathbb{E}[\mathbf{X}\mathbf{X}^\top]) = 0.4167 \geq \kappa_1$. The lower bound is within 10% of the truth.
- Scenario C: $\lambda_{\min}(\Sigma) = 4 \times 10^{-4}, < 10^{-4}, < 10^{-4}$ for $J = 2, 3, 4$.
- (iv) Over 200 replications: full column rank in every replication at $n = 200$ (smallest $\lambda_{\min}(\hat{\Sigma})$ equal to 0.047 in A and 0.081 in B); median $\lambda_{\min}(\hat{\Sigma})/\lambda_{\min}(\Sigma)$ equal to 0.40 (A) and 0.51 (B) at $n = 200$, and 0.84 and 0.96 at $n = 3200$, where $\lambda_{\min}(\hat{\Sigma}) \geq \lambda_{\min}(\Sigma)/2$ in all replications; median $\|\hat{\Sigma} - \Sigma\|_{\text{op}}$ falls from 0.83 to 0.21 (A) and from 0.67 to 0.16 (B), a factor 4.1 and 4.2 for a sixteenfold n ; $\max_i \|\mathbf{Z}_i\|_2^2 \leq 67.3 \leq R_J = 93.6$.
- Compatibility on a support of size 7 (the two unpenalized columns and five random wavelet columns), computed exactly by 2^7 quadratic programs (one per sign pattern of δ_S): population $\phi_0^2(S; \Sigma) = 0.368$ (A) and 0.495 (B), both above $\lambda_{\min}(\Sigma)$; at $n = 200$, over 20 replications, median 0.240 (A) and 0.363 (B), minimum 0.182 and 0.282, always above $\lambda_{\min}(\hat{\Sigma})$.

References

- Bühlmann, P. and van de Geer, S. (2011). *Statistics for High-Dimensional Data*. Springer. (bühlmann2011statistics, verified in L1.) The numbering was checked in L3: Corollary 6.8 is on p. 152, in Section 6.12, and states that $32s \|\Sigma_1 - \Sigma_0\|_\infty / \phi_0^2(S; \Sigma_0) \leq 1$ transfers the compatibility condition from Σ_0 to Σ_1 .

- Cohen, A., Daubechies, I. and Vial, P. (1993). Wavelets on the interval and fast wavelet transforms. *Applied and Computational Harmonic Analysis* 1, 54–81. (`cohen1993wavelets`, verified in L1.)
- Tropp, J. A. (2012). User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics* 12, 389–434. (`tropp2012user`, copied from the WALL bibliography and verified in L3.)